

Group 3 - Space Complexity

Members:

**Kristy Mok, Ben Chalk, Corina Rivero Montiel, Euan Faller, Joel Cutler,
Luke Callen, Sam Ade Fowodu**

User Evaluation Report

- a) We recruited 7 participants from our ENG1 cohort, as they represent the target audience for the game. This ensured relevance, availability, and diversity within a shared university context. Each participant signed an informed consent form to meet ethical standards. We used the following tools for efficient data collection:

- Google Forms: To gather structured feedback and task completion data.
- Observation Notes: To document user actions, errors, and hesitations.

These tools were chosen for their accessibility, ease of use, and ability to provide both quantitative and qualitative data.

The procedure we followed for the user evaluation was the following:

1. Introduction: Explained the evaluation process and emphasized that the system, not the user, was being tested.
2. Tasks: Participants completed four tasks (placing buildings, handling events, and achieving satisfaction goals).
3. Think-Aloud Protocol: Users verbalized their thoughts while performing tasks, helping us understand their mental models.
4. Observation: We noted user actions, delays, and errors using a predefined template.
5. Post-Task Questionnaire: Collected feedback on ease of use, confusion points, and improvement suggestions via Google Forms.
6. Rationale: This method provided comprehensive insights through:
 - Think-Aloud: Understanding user thought processes.
 - Observation: Identifying real-time issues.
 - Google Forms: Structured feedback collection.

b)

Problem ID	Description	Severity	Frequency	User Comments	Proposed Solution
P1	Difficulty understanding how satisfaction is calculated	Major	4/7 users	"How do you reach a higher score?" and "The satisfaction number is really confusing."	Change the satisfaction score to a percentage number, where 100% is clearly the maximum and 0% is the minimum.
P2	Difficulty distinguishing between building types.	Major	3/7 users	"Wasn't sure which building was for studying or eating."	Add labels to the sprites depending on category of buildings (Accommodation, Lecture Hall, etc)
P3	Lack of visible pause	Critical	4/7 users	"Pause is not very clear."	Add a visible pause button

	button			and "Add a pause button."	to the game or specify that to pause the button you have to press the esc key.
P4	Achievements are unclear and difficult to understand.	Minor	4/7 users	"Didn't know what the badges were for." and "Achievements were confusing."	Add descriptions for achievements, a label for the badges specifying that those are the achievements and instructions on how to achieve them.
P5	Accidental deletion of buildings due to unclear control/ Not knowing how to demolish a building	Minor	2/7 users	"Right-clicking deletes a building by accident." and "How do you demolish a building?"	Add a button for demolishing buildings and a confirmation prompt for building deletion.
P6	Ending menu is easy to accidentally exit the game	Minor	1/7	"The menu popped up, and I accidentally exited it."	Require name input before allowing exit of the game.
P7	Difficulty balancing resources and understanding building costs and differences between buildings of same category	Major	2/7	"Balance is not very clear, not sure how much buildings cost" and "I'm not sure the difference between buildings of same"	Display building costs and descriptions more prominently

				category"	
P8	No instructions displayed at the start of the game	Critical	5/7 users	"Instructions at the start would make the game easier to understand"	Add a tutorial or instructions screen at the start of the game.

Severity Scale:

- **Critical:** Prevents task completion.
- **Major:** Significantly hinders task performance.
- **Minor:** Causes inconvenience or confusion but doesn't prevent task completion.
- **Cosmetic:** Aesthetic issue with minimal impact on usability.